

PE-EC702B — Digital Image and Video Processing

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Module 4 Exam Notes: Image Segmentation

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1. Detection of Discontinuities

Image segmentation is based on one of two basic properties of intensity values: discontinuity (partitioning based on abrupt changes) or similarity (partitioning into homogeneous regions).

Edge and Line Detection

Edge Detection: The process of locating boundaries where intensity changes rapidly. These are classified as step edges (ideal abrupt transitions) or ramp edges (gradual, practical transitions due to blur).

1.1 Canny Edge Detection Algorithm

The Canny detector is considered the optimal edge detector due to its low error rate, precise edge localization, and single response constraint.

1. **Noise Reduction:** Smooth the image with a 2D Gaussian filter to eliminate high-frequency noise.
2. **Gradient Computation:** Compute the local gradient magnitude $M(x, y)$ and direction $\alpha(x, y)$ using Sobel or Prewitt masks.
3. **Non-Maximum Suppression (NMS):** Thin the thick edges by checking if the gradient magnitude at a pixel is a local maximum in the direction of the gradient. If not, set it to zero.
4. **Hysteresis Thresholding:** Use two thresholds, T_{high} and T_{low} :
 - Pixels $> T_{\text{high}}$ are marked as strong edges (always kept).
 - Pixels $< T_{\text{low}}$ are suppressed.
 - Pixels in-between are marked as weak edges and kept only if they are connected to a strong edge.

2. Edge Linking and Boundary Detection

Since noise and lighting can cause breaks in detected edges, edge linking is used to group pixels into meaningful boundaries.

2.1 Hough Transform

The Hough Transform groups edge pixels by determining if they lie on a curve of specified shape (typically a straight line).

Hough Transform Formulations

A straight line in Cartesian coordinates is $y = mx + c$.

In Hough parameter space, every point (x, y) maps to a line in (m, c) space:

$$c = -xm + y \quad (1)$$

However, vertical lines have infinite slope ($m \rightarrow \infty$). To resolve this, we use the polar representation:

$$\rho = x \cos \theta + y \sin \theta \quad (2)$$

where ρ is the perpendicular distance from the origin to the line, and θ is the angle of the normal vector relative to the x -axis.

Algorithm:

1. Quantize the parameter space (ρ, θ) into an accumulator grid.
2. For each edge pixel (x_i, y_i) in the image, vary θ over the range $[-90^\circ, 90^\circ]$ and compute the corresponding ρ .
3. Increment the accumulator cell: $A(\rho, \theta) = A(\rho, \theta) + 1$.
4. Local maxima (peaks) in the accumulator grid indicate lines in the image.

3. Thresholding

Thresholding converts a grayscale image $f(x, y)$ into a binary image $g(x, y)$ based on intensity.

3.1 Global vs. Local Adaptive Thresholding

- **Global Thresholding:** A single threshold T is used for the entire image:

$$g(x, y) = \begin{cases} 1 & \text{if } f(x, y) > T \\ 0 & \text{if } f(x, y) \leq T \end{cases} \quad (3)$$

Only effective if the image has a bimodal histogram and uniform illumination.

- **Local/Adaptive Thresholding:** The threshold $T(x, y)$ changes dynamically across the image based on local neighborhood statistics (e.g., local mean or local standard deviation), resolving non-uniform illumination.

3.2 Otsu's Optimum Thresholding Method

Otsu's method automatically selects the optimal global threshold T by maximizing the **between-class variance** σ_B^2 of the two pixel classes (background C_1 and foreground C_2).

Otsu's Formulation

Let $P_1(T)$ and $P_2(T)$ be the probabilities of the two classes separated by threshold T , and let $\mu_1(T)$ and $\mu_2(T)$ be their mean intensities.

The between-class variance $\sigma_B^2(T)$ is defined as:

$$\sigma_B^2(T) = P_1(T)P_2(T) [\mu_1(T) - \mu_2(T)]^2 \quad (4)$$

We evaluate $\sigma_B^2(T)$ for all possible gray levels $T \in [0, L-1]$. The optimum threshold T^* is the value that maximizes this variance:

$$T^* = \arg \max_{0 \leq T \leq L-1} \sigma_B^2(T) \quad (5)$$

4. Region-Based Segmentation

Region-based segmentation groups pixels directly into regions based on similarity.

- **Region Growing:** Starts with seed points and appends adjacent pixels to the region if they satisfy homogeneity criteria (e.g., color, texture, intensity).
- **Region Splitting and Merging:**
 - Start with the entire image region R .
 - If R does not satisfy the homogeneity criterion $P(R) = \text{True}$ (e.g., variance too high), split it into four disjoint quadrants.
 - Recursively split non-homogeneous quadrants. This is modeled by a **quadtree** hierarchy.
 - After splitting ceases, merge adjacent quadrants R_i and R_j if their union satisfies the homogeneity criterion ($P(R_i \cup R_j) = \text{True}$).

Quadtree Decomposition for Splitting and Merging

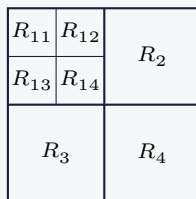
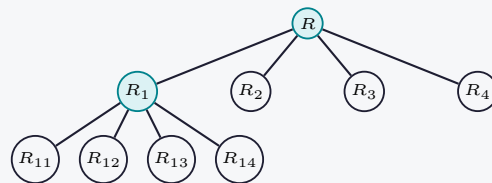


Image Partitions



Quadtree Hierarchy

5. Worked Examples

Hough Transform Line Mapping

Problem: Map the edge point $p(1, 1)$ in Cartesian coordinates into Hough parameter space (ρ, θ) . Specifically, calculate the values of ρ for $\theta = -45^\circ, 0^\circ, 45^\circ$, and 90° .

Solution: We use the polar line equation: $\rho = x \cos \theta + y \sin \theta$. Given $x = 1$ and $y = 1$:

$$\rho(\theta) = 1 \cdot \cos \theta + 1 \cdot \sin \theta = \cos \theta + \sin \theta$$

Let's compute the value of ρ for each requested angle θ :

1. For $\theta = -45^\circ$:

$$\cos(-45^\circ) = \cos(45^\circ) = \frac{1}{\sqrt{2}} \approx 0.707$$

$$\sin(-45^\circ) = -\sin(45^\circ) = -\frac{1}{\sqrt{2}} \approx -0.707$$

$$\rho = 0.707 - 0.707 = 0$$

2. For $\theta = 0^\circ$:

$$\begin{aligned}\cos(0^\circ) &= 1, \quad \sin(0^\circ) = 0 \\ \rho &= 1 + 0 = 1\end{aligned}$$

3. For $\theta = 45^\circ$:

$$\begin{aligned}\cos(45^\circ) &= \frac{1}{\sqrt{2}} \approx 0.707, \quad \sin(45^\circ) = \frac{1}{\sqrt{2}} \approx 0.707 \\ \rho &= 0.707 + 0.707 = 1.414 = \sqrt{2}\end{aligned}$$

4. For $\theta = 90^\circ$:

$$\begin{aligned}\cos(90^\circ) &= 0, \quad \sin(90^\circ) = 1 \\ \rho &= 0 + 1 = 1\end{aligned}$$

Resulting Hough Curves points: The Cartesian point $(1, 1)$ maps to the sinusoidal curve in (ρ, θ) space passing through the coordinates:

$$(-45^\circ, 0), \quad (0^\circ, 1), \quad (45^\circ, \sqrt{2}), \quad (90^\circ, 1)$$

Quick Revision — Module IV

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Key Formulas at a Glance

Polar Line Equation	$\rho = x \cos \theta + y \sin \theta$
Global Thresholding	$g(x, y) = \begin{cases} 1 & \text{if } f(x, y) > T \\ 0 & \text{if } f(x, y) \leq T \end{cases}$
Otsu's Class Probabilities	$P_1(T) = \sum_{i=0}^T p_i, \quad P_2(T) = 1 - P_1(T)$
Otsu's Between-Class Var	$\sigma_B^2(T) = P_1(T)P_2(T)[\mu_1(T) - \mu_2(T)]^2$
Canny Gaussian Smoothing	$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-(x^2+y^2)/2\sigma^2}$
Sobel Gradient Magnitude	$M(x, y) = \sqrt{G_x^2 + G_y^2} \approx G_x + G_y $

One-Line Definitions

- **Hysteresis Thresholding:** A dual-threshold edge linking method that retains weak edges only if connected to strong ones.
- **Hough Transform:** A global parameter mapping technique used to detect straight lines or shapes from broken edge segments.
- **Adaptive Thresholding:** Thresholding where T varies across the image to accommodate local illumination changes.
- **Otsu's Method:** An optimal global thresholding algorithm that automatically chooses T to maximize between-class variance.
- **Quadtree:** A tree data structure in which each internal node has exactly four children, used for image partitioning.
- **Non-Maximum Suppression:** An edge-thinning technique that suppresses non-peak gradient magnitudes along gradient directions.

Frequently Asked / Viva Questions

1. **State the three optimal criteria defined by Canny for edge detection.**
(1) Low error rate (no missed or false edges), (2) Good localization (minimum distance between real and detected edges), and (3) Single response (only one detector response per edge).
2. **Why is polar representation (ρ, θ) preferred over slope-intercept (m, c) in the Hough Transform?**
The slope m approaches infinity for vertical lines, requiring infinite storage size. Polar parameters are bounded: $\theta \in [-90^\circ, 90^\circ]$ and $\rho \in [-D, D]$ (where D is the image diagonal).
3. **How does Otsu's algorithm select the threshold value?**

Frequently Asked / Viva Questions (contd.)

It computes the between-class variance $\sigma_B^2(T)$ for every possible threshold value $T \in [0, L - 1]$, and selects T^* that yields the maximum between-class variance.

4. What is the physical meaning of maximizing between-class variance in thresholding?

Maximizing between-class variance minimizes the spread (within-class variance) of background and foreground pixels, maximizing their statistical separation.

5. Explain the region splitting and merging algorithm.

The entire image is recursively split into quadrants if it fails a homogeneity test (variance too high). After splitting ceases, adjacent quadrants that satisfy the homogeneity test are merged.

6. What is the effect of noise on global thresholding?

Noise distorts the bimodal histogram valleys, causing global thresholding to create false boundaries and merge background and foreground details.

7. What is the function of the Gaussian filter in Canny edge detection?

The Gaussian filter acts as a low-pass filter, smoothing out high-frequency noise which would otherwise cause false edge detections during derivative operations.

8. What are weak edges in Canny edge detection?

Weak edges are edge candidates with gradient magnitudes between T_{low} and T_{high} . They are kept only if they connect to a strong edge.

9. Define region growing.

It is a segmentation procedure that groups pixels into larger regions starting from predefined seed points by adding neighboring pixels that satisfy similarity criteria.

10. How does Sobel gradient calculation differ from Laplacian edge detection?

Sobel uses first derivatives (showing direction and magnitude, robust to noise), while Laplacian uses second derivatives (isotropic, thinned edges, but highly sensitive to noise).

Edge-Based vs. Region-Based Segmentation

Attribute	Edge-Based Segmentation	Region-Based Segmentation
Primary Property	Discontinuity of intensity features	Similarity / Homogeneity of regions
Noise Sensitivity	High (sensitive to noise and gaps)	Low (integrates spatial neighborhood)
Open Boundaries	Yes (requires edge linking)	No (always yields closed regions)